

Integral and Transform Frontiers for the Fabius, Inverse Fabius, and Rvachev Functions

A repository-aware analytic and computational investigation

27 August 2026

Abstract

This report develops a systematic integral calculus for the Fabius function F , Rvachev’s compactly supported function up , and the inverse Fabius function $G = F^{-1}$. It begins with a recursive audit of the current TeX corpus in the **ProveIt** repository, including the main monograph, the consolidated non-formalized frontier document, imported papers, current draft packages, and historical archives. Existing derivative, moment, Fourier-product, Lambert– W , Bell-polynomial, Bernoulli-polynomial, and inverse-endpoint results are treated as baselines rather than reannounced as new.

The principal new results are: a finite dyadic closed form for $\int x^p F(x) dx$ when $p \in \mathbb{N}_0$; an absolutely and locally uniformly convergent dyadic series for arbitrary complex powers, including negative and fractional exponents; two-parameter recurrences for fractional Volterra integrals; complementary Mellin transforms controlled by an entire complex moment function; an entire Newton interpolation of complex moments from rational integer moments; exact local primitives and all-order derivative formulas for the inverse Fabius function; a sharp L^q threshold for G' ; refined inverse-primitive endpoint asymptotics carrying the repository’s log-periodic Lambert phase; Stieltjes, logarithmic-potential, autocorrelation, and order-statistic transforms; and a rational Jacobi continued fraction generated by the Fabius moments. Numerical experiments based on finite Thue–Morse splines independently check the main formulas. Unproved conjectures and a Lean formalization roadmap are included.

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1 Scope, corpus audit, and novelty protocol

1.1 Repository review

The live directory

[https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/
FabiusFunction/docs](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs)

was recursively audited on 27 August 2026. All discoverable TeX paths were inventoried. Current synthesis documents and research drafts were examined through their raw TeX, imported papers were checked for the definitions and results relevant to integration, and historical copies were compared through their content indexes so that superseded duplicates were not counted as independent evidence.

The main baselines were:

- `Fabius_Function_and_Rvachev_Up.tex`, the principal monograph;
- `non-formalized-research-frontiers.tex`, the consolidated frontier synthesis;
- the Lean walkthrough, glossary, and non-elementarity report;
- the imported TeX sources of Arias de Reyna, Haugland, Aistleitner–Hofer–Larcher, Tóth, and related papers;
- the archive index and its historical report families.

The current draft inbox contained the following additional packages:

- | | |
|--|--|
| • <code>Fabius_Arithmetic_Rays_Frontier_Report</code> | • <code>Fabius_Rvachev_Thue_Morse_Frontier_Results</code> |
| • <code>Fabius_Dyadic_Self_Sampling_Frontier_Package</code> | • <code>Spectral_Arithmetic_Pascal_Rvachev_Hierarchy</code> |
| • <code>Fabius_Half_Integer_Spectral_Frontier_Report</code> | • <code>Thue_Morse_Formula_Atlas</code> |
| • <code>Fabius_Inverse_Frontier_Report_Source_and_PDF</code> | • <code>fabius_frontier_dyadic_inverse_barnes_report</code> |
| • <code>Fabius_Newton_Rvachev_Frontier_Report</code> | • <code>fabius_frontier_new_results</code> |
| • <code>Fabius_Rvachev_Frontier_Report-2</code> | • <code>fabius_frontier_results</code> |
| • <code>Fabius_Rvachev_Frontier_Report</code> | • <code>fabius_frontier_results_bundle</code> |
| • <code>Fabius_Rvachev_New_Frontiers</code> | • <code>fabius_frontier_spectral_endpoint_report_bundle</code> |
| | • <code>rvachev_up_fourier_decay</code> |

The primitive identity $F(x/2) = \int_0^x F(t) dt$ is implicit in the audited spectral hierarchy and therefore is used here as a baseline. Likewise, the repository already contains extensive work on integer moments, dyadic values, Fourier sinc products, endpoint asymptotics, inverse asymptotics, and Bell-polynomial reversion. The novelty claims below are deliberately narrower: “new” means not located explicitly in this audited corpus or in the targeted external literature search, not an absolute historical-priority claim.

1.2 Main questions addressed

The report studies:

- normalized and indefinite antiderivatives of F , up , $1 - F$, $F(1 - x)$, and G ;
- polynomial, negative, fractional, complex, exponential, logarithmic, and resolvent weights;
- integer and fractional repeated integration, and all-order differentiation of the inverse;
- Mellin, Laplace, Fourier, Stieltjes/Cauchy, logarithmic-potential, and correlation transforms;
- products and compositions such as $F^a(1 - F)^b$, $\text{up}(2G - 1)^r$, and shifted products of up ;
- exact arithmetic consequences, numerical verification, conjectures, and formalization targets.

2 Definitions and probabilistic normalization

2.1 The bounded Fabius function and Rvachev's function

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius distribution function, with

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (1)$$

and, on $0 \leq x \leq 1/2$,

$$F'(x) = 2F(2x). \quad (2)$$

It obeys

$$F(x) + F(1 - x) = 1, \quad 0 \leq x \leq 1. \quad (3)$$

Rvachev's function is

$$\text{up}(u) = \begin{cases} F(1 - |u|), & |u| \leq 1, \\ 0, & |u| > 1. \end{cases} \quad (4)$$

It is even, nonnegative, C^∞ , supported on $[-1, 1]$, and normalized by $\int_{-1}^1 \text{up}(u) \, du = 1$. The two functions satisfy

$$F'(x) = 2 \text{up}(2x - 1), \quad 0 < x < 1, \quad (5)$$

and, on $[0, 1]$,

$$\text{up}(x) = 1 - F(x) = F(1 - x). \quad (6)$$

The repository also uses a signed global extension $\mathcal{F}$ that agrees with F on $[0, 1]$ and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x), \quad x \in \mathbb{R}. \quad (7)$$

It is indispensable when derivatives move the argument outside $[0, 1]$.

2.2 Random-series model

Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ and set

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}. \quad (8)$$

Then F is the distribution function of X . If $V_j = 2U_j - 1$ and

$$Y = 2X - 1 = \sum_{j=1}^{\infty} \frac{V_j}{2^j}, \quad (9)$$

then Y has density up . In particular,

$$\text{up}(t) = \mathbb{P}(X > t), \quad 0 \leq t \leq 1. \quad (10)$$

This survival-function interpretation is the quickest route to many definite integrals.

Let

$$D(z) = \mathbb{E}X^z, \quad d_n = D(n). \quad (11)$$

Because the density is infinitely flat at 0, all negative moments exist and D extends to an entire function; this is proved in [theorem 7.1](#).

2.3 Endpoint flatness and the inverse

Every derivative of F vanishes at 0, every derivative of $1 - F$ vanishes at 1, and every derivative of $\text{up} - 1$ vanishes at 0. Thus, for each $M \in \mathbb{N}$,

$$F(x) = O(x^M), \quad F'(x) = O(x^M) \quad (x \downarrow 0). \quad (12)$$

These bounds justify complex powers, repeated integration by parts, and normal convergence arguments that would fail for an ordinary density with algebraic endpoint behavior.

The restriction of F to $[0, 1]$ is strictly increasing. Its inverse

$$G : (0, 1) \rightarrow (0, 1), \quad G = F^{-1}, \quad (13)$$

obeys

$$G(1 - y) = 1 - G(y). \quad (14)$$

3 Three master integration principles

Theorem 3.1 (CDF–survival integration). *Let H be absolutely continuous on $[0, 1]$ and suppose the following integrals exist. Then*

$$\int_0^1 H'(t) \text{up}(t) \, dt = \mathbb{E}[H(X) - H(0)], \quad (15)$$

$$\int_0^1 H'(t) F(t) \, dt = H(1) - \mathbb{E}H(X). \quad (16)$$

Proof. For (15), use (10) and Fubini:

$$\int_0^1 H'(t) \mathbb{P}(X > t) \, dt = \mathbb{E} \int_0^X H'(t) \, dt.$$

Equation (16) follows either similarly or by adding the two formulas and using $F + \text{up} = 1$. $\square$

Theorem 3.2 (Quantile push-forward). *For every Borel function Φ for which either side is absolutely integrable,*

$$\int_0^1 \Phi(G(y)) \, dy = \int_0^1 \Phi(x) \, dF(x) = \mathbb{E}\Phi(X). \quad (17)$$

Proof. This is the probability-integral transform, or directly the substitution $y = F(x)$. $\square$

Theorem 3.3 (Universal beta family). *For $\Re a > -1$ and $\Re b > -1$,*

$$\int_0^1 F(x)^a (1 - F(x))^b F'(x) dx = B(a + 1, b + 1). \quad (18)$$

Equivalently,

$$\int_0^1 F(x)^a \operatorname{up}(x)^b \operatorname{up}(2x - 1) dx = \frac{1}{2} B(a + 1, b + 1). \quad (19)$$

Proof. Put $y = F(x)$ and use (5). $\square$

The point of theorems 3.1 and 3.2 is not merely probabilistic elegance. They convert any tractable ordinary antiderivative H into an exact Fabius or Rvachev integral. Monomials give moments, exponentials give Laplace products, logarithms give entropy and logarithmic potentials, and resolvents give Stieltjes transforms.

4 Integer-order primitives and derivative hierarchies

4.1 Normalized primitives of F

From (2),

$$\frac{d}{dx} F(x/2) = F(x), \quad 0 \leq x \leq 1. \quad (20)$$

Thus $F(x/2)$ is the primitive normalized to vanish at $x = 0$.

For $n \in \mathbb{N}$, write

$$I_0^n f(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt. \quad (21)$$

Theorem 4.1 (Exact n th primitive of F). *For $n \in \mathbb{N}$ and $0 \leq x \leq 1$,*

$$\boxed{I_0^n F(x) = 2^{\binom{n}{2}} F\left(\frac{x}{2^n}\right)}. \quad (22)$$

Every other n th antiderivative differs from the right side by a polynomial of degree at most $n - 1$.

Proof. The case $n = 1$ is (20). If the formula holds at order n , then

$$I_0^{n+1} F(x) = 2^{\binom{n}{2}} \int_0^x F(t/2^n) dt = 2^{\binom{n}{2} + n} F(x/2^{n+1}),$$

which is the required triangular exponent. $\square$

For the bounded extension and $x \geq 1$, the n th derivative is 1, so matching derivatives at $x = 1$ gives

$$I_0^n F(x) = \frac{(x-1)^n}{n!} + \sum_{k=0}^{n-1} \frac{2^{\binom{n-k}{2}} F(2^{-(n-k)})}{k!} (x-1)^k. \quad (23)$$

For the signed global extension, a single formula is valid on all of $\mathbb{R}$:

$$\frac{d^n}{dx^n} \left[2^{\binom{n}{2}} \mathcal{F}(x/2^n) \right] = \mathcal{F}(x). \quad (24)$$

4.2 Derivatives and primitives of up

The global CDF of the centered density is

$$\int_{-\infty}^x \text{up}(t) dt = F\left(\frac{x+1}{2}\right). \quad (25)$$

Consequently:

Theorem 4.2 (Left n th primitive of up). *For $n \in \mathbb{N}$ and $x \leq 1$,*

$$\boxed{\frac{1}{(n-1)!} \int_{-\infty}^x (x-t)^{n-1} \text{up}(t) dt = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right).} \quad (26)$$

For $x \geq 1$, it becomes the moment polynomial

$$\sum_{k=0}^{n-1} \frac{(-1)^k \mu_k}{k!(n-1-k)!} x^{n-1-k}, \quad \mu_k = \int_{-1}^1 t^k \text{up}(t) dt. \quad (27)$$

Proof. Equation (25) is the case $n = 1$. Induct as in [theorem 4.1](#), now mapping the lower endpoint -1 to 0 . For $x \geq 1$, expand $(x-t)^{n-1}$ over the compact support. $\square$

On $0 \leq x \leq 1$ one also has the convenient centered primitive

$$\int_0^x \text{up}(t) dt = x - F(x/2) = F\left(\frac{x+1}{2}\right) - \frac{1}{2}. \quad (28)$$

The forward derivative hierarchy is

$$\mathcal{F}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n x), \quad (29)$$

and, for $-1 \leq x \leq 1$,

$$\text{up}^{(n)}(x) = 2^{\binom{n+1}{2}} \mathcal{F}(2^n(x+1)). \quad (30)$$

The antiderivative hierarchy (22) is therefore the exact inverse triangular scaling of the derivative hierarchy.

5 Dyadic antiderivatives with polynomial, fractional, and complex weights

This section contains the main antiderivative results of the report. The elementary dilation primitive in [equation \(20\)](#) becomes substantially more useful after one allows a monomial weight and then analytically continues the exponent.

5.1 A finite closed form for nonnegative integer powers

For $0 < x \leq 1$ and $\nu \in \mathbb{C}$ put

$$A_\nu(x) = \int_0^x t^\nu F(t) dt, \quad t^\nu = e^{\nu \log t}. \quad (31)$$

Endpoint flatness makes this integral locally uniformly convergent for every $\nu \in \mathbb{C}$; hence $A_\nu(x)$ is entire as a function of ν .

Lemma 5.1 (One-step dyadic reduction). *For $0 < x \leq 1$ and every $\nu \in \mathbb{C}$,*

$$A_\nu(x) = x^\nu F(x/2) - \nu 2^\nu A_{\nu-1}(x/2). \quad (32)$$

Proof. On $[0, 1]$, $\frac{d}{dt}F(t/2) = F(t)$. Integration by parts gives

$$A_\nu(x) = [t^\nu F(t/2)]_0^x - \nu \int_0^x t^{\nu-1} F(t/2) dt.$$

The lower boundary vanishes for all complex ν because F is flat at 0. In the remaining integral put $t = 2u$. $\square$

Theorem 5.2 (Closed monomial antiderivative). *For every $p \in \mathbb{N}_0$ and $0 \leq x \leq 1$,*

$$\boxed{\int_0^x t^p F(t) dt = \sum_{j=0}^p (-1)^j \frac{p!}{(p-j)!} 2^{\binom{j+1}{2}} x^{p-j} F\left(\frac{x}{2^{j+1}}\right)}. \quad (33)$$

In particular,

$$\int_0^x t F(t) dt = x F(x/2) - 2 F(x/4), \quad (34)$$

$$\int_0^x t^2 F(t) dt = x^2 F(x/2) - 4x F(x/4) + 16 F(x/8), \quad (35)$$

$$\int_0^x t^3 F(t) dt = x^3 F(x/2) - 6x^2 F(x/4) + 48x F(x/8) - 384 F(x/16). \quad (36)$$

For $x \geq 1$, the bounded extension is obtained by adding the elementary tail

$$A_\nu(x) = A_\nu(1) + \begin{cases} \frac{x^{\nu+1} - 1}{\nu + 1}, & \nu \neq -1, \\ \log x, & \nu = -1, \end{cases} \quad (37)$$

where the first quotient has a removable singularity at $\nu = -1$.

Proof. Iterate (32). At the j th step the falling factorial contributes $p!/(p-j)!$. The powers of 2 telescope to

$$2^{jp-j(j-1)/2} 2^{-j(p-j)} = 2^{j(j+1)/2}.$$

The process terminates because $p^{\overline{p+1}} = 0$. Formula (37) follows from $F = 1$ on $[1, \infty)$. $\square$

Remark 5.3. At $x = 1$, integration by parts against the probability measure dF gives

$$\int_0^1 t^p F(t) dt = \frac{1 - d_{p+1}}{p+1}, \quad \int_0^1 t^p \text{up}(t) dt = \frac{d_{p+1}}{p+1}. \quad (38)$$

Since the integer moments d_n are rational, all these definite integrals are rational.

5.2 A convergent dyadic series for arbitrary complex powers

The recurrence does not terminate for a nonintegral exponent, but endpoint flatness converts it into a genuinely convergent series rather than merely a formal asymptotic expansion.

Theorem 5.4 (Complex-weight dyadic antiderivative series). *Fix $0 < x \leq 1$. For every $\nu \in \mathbb{C}$,*

$$A_\nu(x) = \sum_{j=0}^{\infty} (-1)^j \nu^{\bar{j}} 2^{\binom{j+1}{2}} x^{\nu-j} F\left(\frac{x}{2^{j+1}}\right). \quad (39)$$

The series converges absolutely and locally uniformly in ν . After N terms the exact remainder is

$$R_N = (-1)^N \nu^{\bar{N}} 2^{N\bar{N} - \binom{N}{2}} A_{\nu-N}(x/2^N). \quad (40)$$

Consequently the sum is entire in ν and may be differentiated termwise on compact subsets of the ν -plane.

Proof. Iterating (32) N times gives the displayed partial sum and (40). It remains to prove $R_N \rightarrow 0$. Flatness means that for every $M \in \mathbb{N}$ there is C_M such that $0 \leq F(v) \leq C_M v^M$ on $[0, 1]$. From theorem 4.1,

$$\begin{aligned} 2^{\binom{N}{2}} F(u/2^N) &= \frac{1}{(N-1)!} \int_0^u (u-v)^{N-1} F(v) \, dv \\ &\leq C_M \frac{M!}{(N+M)!} u^{N+M}. \end{aligned} \quad (41)$$

After changing variables in (40),

$$R_N = (-1)^N \nu^{\bar{N}} \int_0^x u^{\nu-N} [2^{\binom{N}{2}} F(u/2^N)] \, du.$$

For ν in a fixed compact set, choose M so that $\Re \nu + M > -1$ uniformly. Bound (41) then gives

$$|R_N| \leq C_{M,K,x} \frac{|\nu^{\bar{N}}|}{(N+M)!}.$$

The quotient tends to zero faster than a chosen inverse power of N , locally uniformly in ν ; at nonnegative integral ν it eventually vanishes identically. This also proves normal convergence. $\square$

Corollary 5.5 (Negative powers). *For $m \in \mathbb{N}$ and $0 < x \leq 1$,*

$$\int_0^x \frac{F(t)}{t^m} \, dt = \sum_{j=0}^{\infty} m^{\bar{j}} 2^{\binom{j+1}{2}} x^{-m-j} F\left(\frac{x}{2^{j+1}}\right). \quad (42)$$

Thus every negative-power antiderivative is finite despite its apparently singular algebraic weight.

Proof. Insert $\nu = -m$ in (39) and use $-m^{\bar{j}} = (-1)^j m^{\bar{j}}$. $\square$

Corollary 5.6 (Weights against up). *For $0 < x \leq 1$ and initially $\Re \nu > -1$,*

$$\int_0^x t^\nu \operatorname{up}(t) \, dt = \frac{x^{\nu+1}}{\nu+1} - A_\nu(x). \quad (43)$$

The right-hand side is the meromorphic continuation in ν , with a single simple pole at $\nu = -1$ of residue 1. The regularized integral

$$\int_0^x t^\nu (\operatorname{up}(t) - 1) \, dt = -A_\nu(x) \quad (44)$$

continues to an entire function of ν .

Frontier contribution. Equations (39) and (42), including normal convergence for arbitrary complex exponent, were not located explicitly in the audited corpus. They supply the requested series form for fractional and negative powers without expanding the nowhere-analytic function F in an ordinary Taylor series.

5.3 Repeated weighted antiderivatives

Corollary 5.7 (Integer-order Volterra primitives with a complex monomial). *For $n \in \mathbb{N}$, $0 < x \leq 1$, and $\nu \in \mathbb{C}$,*

$$I_0^n[t^\nu F(t)](x) = \frac{1}{(n-1)!} \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} x^{n-1-r} A_{\nu+r}(x). \quad (45)$$

For $\nu = p \in \mathbb{N}_0$, substituting (33) gives a finite double sum of dyadically scaled values of F ; for general ν , substituting (39) gives a normally convergent series.

Proof. Expand $(x-t)^{n-1}$ in the defining Volterra integral and collect $A_{\nu+r}(x)$. $\square$

6 Fractional Volterra integration and a two-parameter dyadic recurrence

Definition 6.1. For $\Re \alpha > 0$ and $0 < x \leq 1$, the left Riemann–Liouville integral is

$$(I_0^\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt. \quad (46)$$

Theorem 6.2 (Fractional dilation recurrence). *For $\Re \alpha > 1$ and $0 < x \leq 1$,*

$$I_0^\alpha F(x) = 2^{\alpha-1} I_0^{\alpha-1} F(x/2). \quad (47)$$

More generally, if $m \in \mathbb{N}$ and $m < \Re \alpha$,

$$I_0^\alpha F(x) = 2^{m\alpha-m(m+1)/2} I_0^{\alpha-m} F(x/2^m). \quad (48)$$

At $\alpha = n \in \mathbb{N}$, taking $m = n-1$ recovers (22).

Proof. Substitute $t = 2u$ and use $F(2u) = F'(u)/2$:

$$I_0^\alpha F(x) = \frac{2^{\alpha-1}}{\Gamma(\alpha)} \int_0^{x/2} (x/2-u)^{\alpha-1} F'(u) du.$$

The boundary terms vanish, and one integration by parts yields (47). Iteration gives (48). $\square$

For a weighted kernel define

$$J_{\alpha,\nu}(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} t^\nu F(t) dt. \quad (49)$$

Again, endpoint flatness makes $J_{\alpha,\nu}$ entire in ν for each fixed α with $\Re \alpha > 0$.

Theorem 6.3 (Two-parameter fractional recurrence). *For $0 < x \leq 1$, $\Re \alpha > 1$, and $\nu \in \mathbb{C}$,*

$$J_{\alpha,\nu}(x) = 2^{\alpha+\nu-1} [J_{\alpha-1,\nu}(x/2) - \nu J_{\alpha,\nu-1}(x/2)]. \quad (50)$$

The relation reduces the pair (α, ν) along a two-dimensional dyadic lattice. When ν is a nonnegative integer, all branches that lower ν terminate; when α is an integer, the remaining quantities reduce to theorem 5.7.

Proof. After $t = 2u$ and $F(2u) = F'(u)/2$,

$$J_{\alpha,\nu}(x) = \frac{2^{\alpha+\nu-1}}{\Gamma(\alpha)} \int_0^{x/2} (x/2 - u)^{\alpha-1} u^\nu F'(u) du.$$

Integrating by parts differentiates the product of the power and Volterra kernel. Its two terms are precisely $J_{\alpha-1,\nu}$ and $-\nu J_{\alpha,\nu-1}$. $\square$

Remark 6.4 (Fractional derivatives). For $m - 1 < \Re \alpha < m$, the Riemann–Liouville derivative $D_0^\alpha F = D^m I_0^{m-\alpha} F$ can be reduced by (47) before differentiation. Because every derivative of F at 0 vanishes, Caputo and Riemann–Liouville derivatives coincide on F for positive nonintegral orders. A useful formalization target is to prove this statement in a function space where the dyadic dilation operator and the fractional Volterra semigroup are both bounded.

7 Mellin transforms, logarithmic moments, and Newton interpolation

7.1 The entire complex moment function

Proposition 7.1 (Entire moment interpolation). *The function*

$$D(z) = \int_0^1 x^z dF(x) = \int_0^1 x^z F'(x) dx$$

is entire on $\mathbb{C}$. Its derivatives are

$$D^{(m)}(z) = \int_0^1 x^z (\log x)^m dF(x). \quad (51)$$

The integer moments satisfy $d_0 = 1$ and

$$(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{d_k}{n - k + 1}, \quad n \geq 1. \quad (52)$$

In particular all d_n are rational.

Proof. The density F' is flat at 0. On each compact set of z -values, a sufficiently high power bound for F' dominates $x^{\Re z} |\log x|^m$ near zero, so differentiation under the integral is valid to every order. For (52), use the distributional fixed point

$$X \stackrel{d}{=} \frac{U + X'}{2},$$

where U is uniform on $[0, 1]$ and X' is an independent copy of X . Expanding $(U + X')^n$ and isolating the $k = n$ term yields the recurrence. $\square$

The first few moments are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \dots \quad (53)$$

7.2 Mellin transforms of F and up

Theorem 7.2 (Complementary Mellin transforms). *For $z \in \mathbb{C}$,*

$$\mathcal{M}_F(z) := \int_0^1 x^{z-1} F(x) \, dx = \frac{1 - D(z)}{z}, \quad (54)$$

where the quotient has a removable singularity at $z = 0$ and is entire. Initially for $\Re z > 0$,

$$\mathcal{M}_{\text{up},+}(z) := \int_0^1 x^{z-1} \text{up}(x) \, dx = \frac{D(z)}{z}. \quad (55)$$

The latter identity is its meromorphic continuation to $\mathbb{C}$; it has only one pole, a simple pole at $z = 0$ with residue 1.

Proof. Stieltjes integration by parts gives

$$D(z) = 1 - z \int_0^1 x^{z-1} F(x) \, dx$$

where first justified in a right half-plane and then continued by [theorem 7.1](#). Equation (55) follows from $\text{up} = 1 - F$ on $[0, 1]$. $\square$

Corollary 7.3 (Logarithmic integral hierarchy). *For $m \in \mathbb{N}_0$,*

$$\int_0^1 \frac{(\log x)^m}{x} F(x) \, dx = -\frac{D^{(m+1)}(0)}{m+1}. \quad (56)$$

Equivalently,

$$\int_0^1 \frac{(\log x)^m}{x} (\text{up}(x) - 1) \, dx = \frac{D^{(m+1)}(0)}{m+1}. \quad (57)$$

In particular,

$$\int_0^1 \frac{F(x)}{x} \, dx = -\mathbb{E} \log X. \quad (58)$$

Proof. Differentiate (54) at $z = 0$ and use the Taylor expansion of D . $\square$

7.3 A Newton series reconstructed from rational moments

At $x = 1$, the dyadic points in (39) can be eliminated by the repository identity

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}). \quad (59)$$

This produces an interpolation formula that reconstructs every complex moment from the rational moments at nonnegative integers.

Corollary 7.4 (Newton–Mellin interpolation). *For every $\nu \in \mathbb{C}$,*

$$\frac{1 - D(\nu + 1)}{\nu + 1} = \sum_{j=0}^{\infty} (-1)^j \nu^j \frac{d_{j+1}}{(j+1)!}, \quad (60)$$

with the left side interpreted removably at $\nu = -1$. Equivalently, for every $z \in \mathbb{C}$,

$$D(z) = 1 - z \sum_{j=0}^{\infty} (-1)^j z^{-1j} \frac{d_{j+1}}{(j+1)!}. \quad (61)$$

The series converges locally uniformly on $\mathbb{C}$.

Proof. Set $x = 1$ in (39), use (59), and combine with (54). Normal convergence was proved in theorem 5.4. It can also be seen directly: endpoint flatness at 1 makes $d_n = O_M(n^{-M})$ for every M , whereas $z - 1^j/j!$ has at most polynomial growth on compact z -sets. $\square$

Corollary 7.5 (A rapidly convergent logarithmic constant).

$$-\mathbb{E} \log X = \int_0^1 \frac{F(x)}{x} dx = \sum_{n=1}^{\infty} \frac{d_n}{n} = 0.758239999 \dots \quad (62)$$

The decimal is numerical, while the series identity is exact.

Proof. Put $z = 0$ in (60); then $(-1)^j - 1^j = j!$. The probabilistic identity $\sum_{n \geq 1} d_n/n = \mathbb{E}[-\log(1 - X)] = \mathbb{E}[-\log X]$ provides an independent check using symmetry. $\square$

Frontier contribution. Formula (61) is an entire Newton interpolation of a non-elementary complex moment function from rational data. It creates a direct bridge between the dyadic Fabius values, the moment arithmetic, and complex Mellin analysis.

7.4 A complex fixed-point equation

Proposition 7.6 (Moment fixed point). *For all $z \in \mathbb{C}$, by removable continuation at $z = -1$,*

$$D(z) = \frac{2^{-z}}{z+1} \mathbb{E} \left[(1+X)^{z+1} - X^{z+1} \right]. \quad (63)$$

At $z = -1$ this reads

$$D(-1) = 2\mathbb{E} \log \frac{1+X}{X}. \quad (64)$$

Proof. Condition on X' in $X \stackrel{d}{=} (U + X')/2$ and integrate $(u + X')^z$ over $0 \leq u \leq 1$. $\square$

One may expand $(1+X)^{z+1}$ in a generalized binomial series. The rapid decay of d_n then gives, initially in a standard convergence region and subsequently by continuation,

$$D(z) = \frac{2^{-z}}{z+1} \left[\sum_{n=0}^{\infty} \binom{z+1}{n} d_n - D(z+1) \right]. \quad (65)$$

This identity is a promising starting point for analytic difference equations in the Mellin variable.

7.5 Mellin–Laplace bridges

Let $B(s) = \mathbb{E}e^{-sX}$, developed in the next section. For $\Re z < 0$,

$$D(z) = \frac{1}{\Gamma(-z)} \int_0^\infty s^{-z-1} B(s) \, ds. \quad (66)$$

More generally, if $m < \Re z < m + 1$ with $m \in \mathbb{N}_0$, then

$$D(z) = \frac{1}{\Gamma(-z)} \int_0^\infty s^{-z-1} \left[B(s) - \sum_{k=0}^m \frac{(-s)^k d_k}{k!} \right] \, ds. \quad (67)$$

The subtractions regularize the origin, while $m < \Re z$ controls the polynomial tail at infinity. These formulas convert asymptotics of the infinite product B into complex-moment information and are suitable for saddle-point analysis of negative moments.

8 Laplace, Fourier, and Bell–Bernoulli transforms

8.1 The one-sided Laplace product

Theorem 8.1 (Laplace product and three exact transforms). *For $s \in \mathbb{C}$,*

$$B(s) = \mathbb{E}e^{-sX} = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad (68)$$

with each zero-over-zero factor understood by continuity. The product is entire and satisfies

$$B(s) = \frac{1 - e^{-s/2}}{s/2} B(s/2). \quad (69)$$

Moreover,

$$\int_0^1 e^{-sx} \operatorname{up}(x) \, dx = \frac{1 - B(s)}{s}, \quad (70)$$

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{B(s) - e^{-s}}{s}, \quad (71)$$

$$\int_0^1 e^{-sG(y)} \, dy = B(s). \quad (72)$$

All quotients are interpreted by removable continuation at $s = 0$.

Proof. Independence in (8) gives (68). The first two integral formulas follow from the master identities in [theorem 3.1](#) with an exponential antiderivative. The last is the quantile substitution (17). $\square$

For every $p \in \mathbb{N}_0$, differentiation supplies weighted transforms without any new quadrature:

$$\int_0^1 x^p e^{-sx} \operatorname{up}(x) \, dx = (-1)^p \frac{d^p}{ds^p} \frac{1 - B(s)}{s}, \quad (73)$$

$$\int_0^1 x^p e^{-sx} F(x) \, dx = (-1)^p \frac{d^p}{ds^p} \frac{B(s) - e^{-s}}{s}, \quad (74)$$

$$\int_0^1 G(y)^p e^{-sG(y)} \, dy = (-1)^p B^{(p)}(s). \quad (75)$$

For a complex power, Mellin inversion or the series of [theorem 5.4](#) provides the corresponding continuation.

8.2 Bernoulli cumulants and Bell-polynomial derivatives

Let B_{2m} denote the Bernoulli numbers with $B_2 = 1/6$. Near the origin,

$$\log B(s) = -\frac{s}{2} + \sum_{m=1}^{\infty} \frac{B_{2m}}{2m(2m)!(2^{2m}-1)} s^{2m}, \quad |s| < 4\pi. \quad (76)$$

Thus the cumulants of X are

$$\kappa_1 = \frac{1}{2}, \quad \kappa_{2m} = \frac{B_{2m}}{2m(2^{2m}-1)}, \quad \kappa_{2m+1} = 0 \quad (m \geq 1). \quad (77)$$

If $\mathcal{B}_n$ is the complete exponential Bell polynomial, then

$$d_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n). \quad (78)$$

At a general transform parameter,

$$B^{(n)}(s) = B(s) \mathcal{B}_n((\log B)'(s), \dots, (\log B)^{(n)}(s)). \quad (79)$$

Equations (75) and (79) therefore express every exponentially damped inverse moment through Bernoulli data and Bell polynomials.

8.3 Bilateral transform of up and the sinc product

Since $Y = 2X - 1$ has density up,

$$\int_{-1}^1 e^{-su} \text{up}(u) \, du = e^s B(2s) = \prod_{j=1}^{\infty} \frac{\sinh(s/2^j)}{s/2^j}. \quad (80)$$

With the angular-frequency convention

$$\phi_Y(\omega) = \int_{-1}^1 e^{i\omega u} \text{up}(u) \, du,$$

one obtains

$$\phi_Y(\omega) = \prod_{j=1}^{\infty} \text{sinc}(\omega/2^j), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (81)$$

In the repository convention $\widehat{f}(\xi) = \int f(x) e^{-2\pi i \xi x} \, dx$, this becomes

$$\widehat{\text{up}}(\xi) = \prod_{n=0}^{\infty} \text{sinc}(\pi \xi / 2^n). \quad (82)$$

Polynomially weighted Fourier transforms follow from

$$\int_{-1}^1 u^n e^{i\omega u} \text{up}(u) \, du = i^{-n} \phi_Y^{(n)}(\omega). \quad (83)$$

Away from the zeros of the product, $\phi_Y^{(n)}$ can again be organized by Bell polynomials applied to the logarithmic derivatives of the sinc factors. At a zero, the entire derivative in (83), rather than a singular logarithmic representation, is the correct object.

9 Integral calculus of the inverse Fabius function

9.1 Global inverse moments and local primitives

The quantile push-forward immediately turns every inverse-Fabius moment into a moment of X .

Theorem 9.1 (Entire inverse-power transform). *For every $z \in \mathbb{C}$,*

$$\int_0^1 G(y)^z dy = D(z). \quad (84)$$

The integral converges for all complex z , including arbitrarily large negative real part.

Proof. Equation (84) is (17) with $\Phi(x) = x^z$. Convergence follows from theorem 7.1; equivalently, $G(y)$ tends to zero more slowly than every positive power of y , so negative powers of G remain integrable at $y = 0$. $\square$

Theorem 9.2 (Exact local inverse-power primitive). *Let $0 \leq y \leq 1$, put $x = G(y)$, and let $p \in \mathbb{C}$. Then*

$$\boxed{\int_0^y G(t)^p dt = yx^p - pA_{p-1}(x).} \quad (85)$$

For $p \in \mathbb{N}$, theorem 5.2 makes the right side a finite sum over $F(x/2^j)$, $1 \leq j \leq p$. For arbitrary complex p , theorem 5.4 gives a normally convergent dyadic series.

Proof. Use $t = F(u)$, so $u = G(t)$ and $dt = F'(u) du$:

$$\int_0^y G(t)^p dt = \int_0^x u^p F'(u) du = yx^p - p \int_0^x u^{p-1} F(u) du.$$

The lower boundary vanishes for every complex p by endpoint flatness. $\square$

For example,

$$\int_0^y G(t) dt = yG(y) - F(G(y)/2), \quad (86)$$

$$\int_0^y G(t)^2 dt = yG(y)^2 - 2G(y)F(G(y)/2) + 4F(G(y)/4), \quad (87)$$

$$\int_0^y G(t)^3 dt = yG(y)^3 - 3G(y)^2 F(G(y)/2) + 12G(y)F(G(y)/4) - 48F(G(y)/8). \quad (88)$$

At $y = 1$, these identities reduce to $\int_0^1 G^p = d_p$.

Corollary 9.3 (A weighted CDF–quantile duality). *For $p \in \mathbb{C}$ and $\Re q > 0$,*

$$\int_0^1 G(y)^p y^{q-1} dy = \frac{1}{q} - \frac{p}{q} \int_0^1 x^{p-1} F(x)^q dx. \quad (89)$$

The endpoint at $x = 0$ causes no restriction on p because $F(x)^q$ is flat whenever $\Re q > 0$.

Proof. Put $y = F(x)$ and integrate $x^p d(F(x)^q)/q$ by parts. $\square$

A second universal identity, useful for inverse derivatives, is

$$\int_0^1 \Phi(G(y))G'(y) dy = \int_0^1 \Phi(x) dx \quad (90)$$

whenever either side is absolutely integrable.

9.2 All derivatives of the inverse

Set $f = F'$. Define differential polynomials $Q_n[f]$ recursively by

$$Q_1 = 1, \quad Q_{n+1} = fQ'_n - (2n-1)f'Q_n. \quad (91)$$

Theorem 9.4 (All-order inverse derivative formula). *If $x = G(y)$, then for every $n \geq 1$,*

$$\boxed{G^{(n)}(y) = \frac{Q_n[f](x)}{f(x)^{2n-1}}.} \quad (92)$$

The first cases are

$$\begin{aligned} G'(y) &= \frac{1}{F'(x)}, \\ G''(y) &= -\frac{F''(x)}{F'(x)^3}, \\ G'''(y) &= \frac{3F''(x)^2 - F'(x)F'''(x)}{F'(x)^5}, \end{aligned} \quad (93)$$

$$G^{(4)}(y) = \frac{-15F''(x)^3 + 10F'(x)F''(x)F'''(x) - (F'(x))^2F^{(4)}(x)}{F'(x)^7}. \quad (94)$$

Using the signed extension,

$$F^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad 0 < x < 1, \quad (95)$$

so every inverse derivative is a rational expression in finitely many dyadically dilated values of $\mathcal{F}$.

Proof. The differentiation operator in the inverse coordinate is $\frac{d}{dy} = f(x)^{-1} \frac{d}{dx}$. Assuming (92) at order n and differentiating gives

$$\frac{1}{f} \frac{d}{dx} \left(Q_n f^{-(2n-1)} \right) = \frac{fQ'_n - (2n-1)f'Q_n}{f^{2n+1}},$$

which is exactly (91). □

The recurrence is a compact alternative to a full Faà di Bruno expansion. The Q_n are closely related to the inverse-function Bell polynomials; (91) is often more efficient computationally because it preserves the natural denominator f^{2n-1} .

9.3 Sharp integrability of the first inverse derivative

Theorem 9.5 (L^q threshold for G'). *For $q > 0$,*

$$\int_0^1 |G'(y)|^q dy = \int_0^1 F'(x)^{1-q} dx. \quad (96)$$

Consequently,

$$G' \in L^q(0, 1) \iff 0 < q \leq 1. \quad (97)$$

In particular $G \in W^{1,1}(0, 1)$ but $G \notin W^{1,q}(0, 1)$ for any $q > 1$.

Proof. Substitute $y = F(x)$ in the left side. If $0 < q \leq 1$, the exponent $1 - q$ is nonnegative and F' is bounded. If $q > 1$, flatness gives $F'(x) \leq C_N x^N$ near 0 for every N . Choose N with $N(q - 1) \geq 1$; then $F'(x)^{1-q}$ dominates a nonintegrable multiple of $x^{-N(q-1)}$. $\square$

The complementary density–quantile identity is

$$\int_0^1 G'(y)^{-q} dy = \int_0^1 F'(x)^{q+1} dx, \quad (98)$$

whenever finite. Two notable specializations are

$$\begin{aligned} \int_0^1 \log G'(y) dy &= - \int_0^1 F'(x) \log F'(x) dx \\ &= - \int_{-1}^1 \text{up}(u) \log(2 \text{up}(u)) du, \end{aligned} \quad (99)$$

$$\int_0^1 \text{up}(2G(y) - 1)^r dy = \int_{-1}^1 \text{up}(u)^{r+1} du, \quad r > -1. \quad (100)$$

Formula (99) identifies the mean logarithmic slope of the quantile with the differential entropy of the Fabius density.

10 Endpoint asymptotics of inverse primitives

The audited inverse-function reports derive a Lambert– W endpoint law with a small periodic fluctuation. Write

$$T = \log(1/y), \quad L = \log 2, \quad \rho = \sqrt{\frac{2T}{L}}, \quad \beta = \frac{1}{L} + \frac{1}{2}. \quad (101)$$

Their leading inversion gives

$$G(y) \sim \frac{\sqrt{T}}{e\sqrt{L}} \exp(-\sqrt{2LT}), \quad y \downarrow 0. \quad (102)$$

In particular G is slowly varying at zero: $G(cy)/G(y) \rightarrow 1$ for each fixed $c > 0$.

Let Ψ denote the bounded period-1 phase function in the repository’s refined inverse asymptotic and define

$$a(\rho) = \frac{\Psi'(\rho + \beta) - 1}{L}. \quad (103)$$

The logarithmic elasticity is

$$\epsilon(y) := \frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{a(\rho)}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right). \quad (104)$$

Theorem 10.1 (Refined inverse-power primitive). *Assume the derivative-controlled form of the all-orders inverse phase expansion underlying (104). For each fixed $p \in \mathbb{C}$, as $y \downarrow 0$,*

$$A_{p-1}(G(y)) = yG(y)^p \left[\frac{1}{\rho} + \frac{a(\rho) - p}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right], \quad (105)$$

$$\int_0^y G(t)^p dt = yG(y)^p \left[1 - \frac{p}{\rho} + \frac{p(p - a(\rho))}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right]. \quad (106)$$

In particular,

$$F(G(y)/2) = yG(y) \left[\frac{1}{\rho} + \frac{a(\rho) - 1}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right]. \quad (107)$$

The period-1 ripple survives integration, but first appears one order below the leading correction.

Proof. Putting $u = G(t)$ in $A_{p-1}(G(y))$ gives the exact identity

$$A_{p-1}(G(y)) = \int_0^y G(t)^p \epsilon(t) dt. \quad (108)$$

Let $y = e^{-T}$ and $H(T) = G(e^{-T})^p \epsilon(e^{-T})$. Then

$$A_{p-1}(G(y)) = \int_T^\infty e^{-u} H(u) du = e^{-T} (H(T) + H'(T) + O(H''(T))).$$

The derivative-controlled version of (104) gives

$$H'(T) = G(e^{-T})^p \left[-\frac{p}{\rho^2} + O\left(\frac{\log \rho}{\rho^3}\right) \right], \quad H''(T) = O\left(G(e^{-T})^p \frac{\log \rho}{\rho^3}\right).$$

Substitution proves (105). Formula (106) follows from (85), and (107) is the case $p = 1$ because $A_0(x) = F(x/2)$. $\square$

Without the second-order phase information, Karamata's theorem already gives the robust leading form

$$\int_0^y t^{q-1} G(t)^p dt \sim \frac{y^q G(y)^p}{q}, \quad q > 0, \quad (109)$$

for every fixed real p . Equation (106) resolves the first two corrections in the important case $q = 1$.

11 Stieltjes, Cauchy, and logarithmic transforms

For $w \in \mathbb{C} \setminus [0, 1]$ define the Cauchy–Stieltjes transform

$$S(w) = \mathbb{E} \frac{1}{w - X} = \int_0^1 \frac{dF(x)}{w - x}. \quad (110)$$

Theorem 11.1 (Four equivalent Stieltjes representations). *For $w \notin [0, 1]$,*

$$S(w) = \int_0^1 \frac{dy}{w - G(y)}, \quad (111)$$

$$= \frac{1}{w} + \int_0^1 \frac{\text{up}(t)}{(w - t)^2} dt, \quad (112)$$

$$= \frac{1}{w - 1} - \int_0^1 \frac{F(t)}{(w - t)^2} dt. \quad (113)$$

For $|w| > 1$,

$$S(w) = \sum_{n=0}^{\infty} \frac{d_n}{w^{n+1}}. \quad (114)$$

The transform of Rvachev's centered density is

$$\int_{-1}^1 \frac{\text{up}(u)}{w - u} du = \frac{1}{2} S\left(\frac{w + 1}{2}\right), \quad w \notin [-1, 1]. \quad (115)$$

Proof. Equation (111) is the quantile push-forward. Apply theorem 3.1 to $H(t) = (w - t)^{-1}$ to obtain (112); Stieltjes integration by parts gives (113). Geometric expansion gives (114). Finally use $u = 2x - 1$ and $dF(x) = 2 \operatorname{up}(2x - 1) dx$. $\square$

A primitive in the spectral variable produces logarithmic kernels.

Corollary 11.2 (Logarithmic potentials). *Choose a branch of the logarithm on a simply connected domain avoiding $[0, 1]$. Then*

$$\int_0^1 \frac{F(x)}{w - x} dx = \mathbb{E} \log(w - X) - \log(w - 1), \quad (116)$$

$$\int_0^1 \frac{\operatorname{up}(x)}{w - x} dx = \log w - \mathbb{E} \log(w - X), \quad (117)$$

$$\int_0^1 \log(w - G(y)) dy = \mathbb{E} \log(w - X). \quad (118)$$

Differentiating any one of these equations recovers the corresponding Stieltjes transform.

Proof. Integrate by parts using $\frac{d}{dx} \log(w - x) = -(w - x)^{-1}$ and apply the quantile push-forward to the last formula. $\square$

Boundary values of S encode the Fabius density through the Sokhotski–Plemelj formula. Thus the Cauchy transform packages moment arithmetic outside the interval and reconstructs F' on the cut. Its continued-fraction expansion, determined by the Hankel determinants of (d_n) , is developed below.

12 Products, order statistics, and autocorrelation

12.1 Products of F and $1 - F$

For positive integers m, n , let $A_m = \max(X_1, \dots, X_m)$ and $B_n = \min(X'_1, \dots, X'_n)$, with the two samples independent. Then

$$\int_0^1 F(x)^m (1 - F(x))^n dx = \mathbb{E}(B_n - A_m)_+. \quad (119)$$

Indeed, the integrand is $\mathbb{P}(A_m \leq x < B_n)$, and integration measures the random interval length. Two useful special cases are

$$\int_0^1 F(x)^m dx = 1 - \mathbb{E}A_m, \quad \int_0^1 \operatorname{up}(x)^m dx = \mathbb{E}B_m. \quad (120)$$

Symmetry gives equality of the two integrals.

The first mixed product admits several especially transparent forms.

Theorem 12.1 (A Fabius Gini integral). *Let X' be an independent copy of X . Then*

$$C_F := \int_0^1 F(x)(1 - F(x)) dx = \frac{1}{2} \mathbb{E}|X - X'|, \quad (121)$$

$$= \frac{1}{\pi} \int_0^\infty \frac{1 - |\phi_X(t)|^2}{t^2} dt, \quad (122)$$

where

$$\phi_X(t) = e^{it/2} \prod_{j=1}^{\infty} \text{sinc}(t/2^{j+1}). \quad (123)$$

Equivalently,

$$C_F = \frac{1}{2\pi} \int_0^{\infty} \frac{1 - \phi_Y(t)^2}{t^2} dt. \quad (124)$$

Furthermore,

$$\int_{-1}^1 \text{up}(u)^2 du = 1 - 2C_F, \quad \int_0^1 F(x)^2 dx = \frac{1}{2} - C_F. \quad (125)$$

Numerically,

$$C_F \approx 0.0955632. \quad (126)$$

Proof. The standard one-dimensional Gini identity gives $\mathbb{E}|X - X'| = 2 \int F(1 - F)$. The Fourier representation follows from $|z| = \frac{2}{\pi} \int_0^{\infty} (1 - \cos tz) t^{-2} dt$ and the characteristic function of $X - X'$. Since $X = (Y + 1)/2$, (124) follows by scaling. Finally, on $[0, 1]$ one has $\text{up} = 1 - F$, and symmetry gives $\int F^2 = \int (1 - F)^2$. Expanding $F(1 - F)$ proves (125). $\square$

12.2 The autocorrelation transform

Define

$$R(a) = \int_{\mathbb{R}} \text{up}(x) \text{up}(x - a) dx. \quad (127)$$

Because up is even, $R = \text{up} * \text{up}$; it is an even C^∞ probability density supported on $[-2, 2]$. In angular-frequency convention,

$$R(a) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\omega a} \prod_{j=1}^{\infty} \text{sinc}^2(\omega/2^j) d\omega. \quad (128)$$

If $\mu_n = \int u^n \text{up}(u) du$, then

$$\int_{-2}^2 a^n R(a) da = \sum_{k=0}^n \binom{n}{k} \mu_k \mu_{n-k}. \quad (129)$$

Only even indices survive. More generally, correlations of derivatives are obtained by multiplying the integrand in (128) by powers of $i\omega$. These identities translate questions about products of shifted Rvachev functions into the arithmetic of a squared lacunary sinc product.

13 Numerical experiments

13.1 Finite Thue–Morse spline model

The accompanying program uses the finite random sum

$$Y_n = \sum_{j=1}^n \frac{V_j}{2^j} \quad (130)$$

and its compactly supported spline density. Let $\tau_j = (-1)^{s_2(j)}$, $A_n = 1 - 2^{-n}$, and $(r)_+ = \max(r, 0)$. Then

$$p_n(y) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \tau_j \left(y + A_n - \frac{j}{2^{n-1}} \right)_+^{n-1}, \quad (131)$$

$$P_n(y) = \frac{2^{\binom{n}{2}}}{n!} \sum_{j=0}^{2^n-1} \tau_j \left(y + A_n - \frac{j}{2^{n-1}} \right)_+^n. \quad (132)$$

The finite Fabius CDF is $F_n(x) = P_n(2x - 1)$. Long-double arithmetic and symmetry are used to suppress cancellation. Exact rational moments are generated independently from (52).

13.2 Results and reproducibility

The first table gives the largest residual among several test points for $\int_0^x F_n(t) dt \approx F_n(x/2)$, together with the finite-rank approximation to C_F . The second tests (33) at $x = 0.83$ and rank 10. The third shows convergence of (61) at $z = 0.5 + 0.7i$.

Spline rank	max. primitive residual	$\int_0^1 F_n(1 - F_n)$
6	$1.873e - 05$	0.095552260300
8	$1.170e - 06$	0.095562546323
10	$7.313e - 08$	0.095563189160

p	quadrature	dyadic formula	difference
0	0.330437473102	0.330437457904	$1.520e - 08$
1	0.205934882125	0.205935065265	$-1.831e - 07$
2	0.135359629291	0.135359392297	$2.370e - 07$
3	0.092437619662	0.092438027029	$-4.074e - 07$
5	0.046620893512	0.046622072851	$-1.179e - 06$

Newton terms	approximation error	approximation
5	$8.605e - 03$	$0.601640735 - 0.321223181 i$
10	$1.041e - 03$	$0.598727644 - 0.314110902 i$
20	$6.790e - 05$	$0.598823615 - 0.313135185 i$
40	$2.260e - 06$	$0.598859558 - 0.313079898 i$
60	$2.590e - 07$	$0.598861332 - 0.313078830 i$
100	$1.245e - 07$	$0.598861546 - 0.313078775 i$

The beta integral (18) was also evaluated directly for several nonintegral exponent pairs; residuals were between 4×10^{-16} and 6×10^{-14} . A rank-10, 400-node quantile quadrature checked (72) with errors

$$5.7 \times 10^{-9}, \quad 3.6 \times 10^{-8}, \quad 7.3 \times 10^{-8} \quad (133)$$

for $s = 0.7, 2.5, 5$, respectively. The partial sums $\sum_{n \leq 200} d_n/n$ give 0.758239999524, agreeing with (62).

The finite spline is not forced to obey the infinite-rank dyadic primitive identity exactly; its residual falls by about a factor of 16 whenever the rank increases by 2, consistent with convergence to the limiting law. By contrast, the beta substitution is exact for every smooth finite-rank CDF, so its residual is essentially floating-point roundoff.

The archive contains `numerical_experiments.py`. Running

```
python numerical_experiments.py
```

regenerates `numerical_results.txt` and `numerical_results.tex`. The code documents every normalization, the finite spline formula, quadrature, bisection inversion, and rational-moment recurrence.

14 An arithmetic continued fraction generated by the moments

The Stieltjes transform admits one more exact structure that appears not to have been exploited in the audited corpus. Let P_n be the monic polynomials orthogonal with respect to dF on $[0, 1]$, normalized by $P_{-1} = 0$, $P_0 = 1$.

Proposition 14.1 (Rational Jacobi parameters). *There are positive rational numbers b_n such that*

$$P_{n+1}(x) = \left(x - \frac{1}{2}\right) P_n(x) - b_n P_{n-1}(x), \quad n \geq 0, \quad (134)$$

where the term containing b_0 is absent. In particular,

$$b_1 = \frac{1}{36}, \quad b_2 = \frac{8}{225}. \quad (135)$$

Consequently the Cauchy transform has the Stieltjes J -fraction

$$S(w) = \frac{1}{w - \frac{1}{2} - \frac{b_1}{w - \frac{1}{2} - \frac{b_2}{w - \frac{1}{2} - \ddots}}}. \quad (136)$$

Every finite convergent is a rational function with rational coefficients.

Proof. Gram–Schmidt applied to $1, x, x^2, \dots$ uses only the rational moments d_n , so every coefficient of P_n and every squared norm $h_n = \int P_n^2 dF$ is rational. Positivity of the measure gives $h_n > 0$ and $b_n = h_n/h_{n-1} > 0$. Symmetry under $x \mapsto 1 - x$ implies $P_n(1 - x) = (-1)^n P_n(x)$, hence $\int x P_n(x)^2 dF(x) = h_n/2$ and every diagonal Jacobi coefficient equals $1/2$. The standard resolvent expansion of the Jacobi matrix gives (136). Finally $b_1 = \text{Var}(X) = 1/36$; using the fourth centered moment $19/10800$ gives $b_2 = 8/225$. $\square$

This rational Jacobi matrix is a new arithmetic object attached to the Fabius law. Determining valuations, denominator growth, and the rate at which b_n approaches the universal interval value is a natural continuation of the repository’s dyadic arithmetic program.

15 Conjectures and frontier directions

15.1 Higher inverse derivatives

Let

$$R_n(y) = \frac{y^n G^{(n)}(y)}{G(y)}, \quad \delta = y \frac{d}{dy}. \quad (137)$$

Direct differentiation gives the exact recurrence

$$R_1 = \epsilon, \quad R_{n+1} = (\epsilon - n)R_n + \delta R_n. \quad (138)$$

The first-order elasticity and the all-orders Lambert phase strongly suggest the following.

Conjecture 15.1 (All-order inverse endpoint law). *For every fixed $n \geq 1$,*

$$G^{(n)}(y) \sim (-1)^{n-1} (n-1)! \frac{G(y)}{y^n \rho(y)}, \quad y \downarrow 0. \quad (139)$$

Consequently, for $n \geq 1$ and $q > 0$,

$$G^{(n)} \in L^q(0, 1) \iff 0 < q \leq \frac{1}{n}. \quad (140)$$

The case $n = 1$ is [theorem 9.5](#). At the borderline $q = 1/n$, the residual factor $G(y)^{1/n}/\rho(y)^{1/n}$ should make the logarithmic-coordinate integral convergent.

A proof should follow by establishing derivative-stable error estimates for the all-orders inverse phase, then inducting in (138). This is more delicate than formal differentiation because an undifferentiated asymptotic remainder does not control its derivatives.

15.2 Negative moments and a dual endpoint saddle

The entire function D grows rapidly on the negative real axis. The dyadic endpoint formula, Stirling's approximation, and the forward saddle suggest the coarse scale

Conjecture 15.2 (Negative-moment saddle). *As $r \rightarrow +\infty$,*

$$\log D(-r) = \frac{\log 2}{2} r^2 - r \log r + O(r). \quad (141)$$

After subtracting the complete smooth expansion in r , a bounded period-1 function of the corrected phase $r - \log_2 r + O(1)$ should remain.

This problem is a Mellin-dual counterpart of the small- x and inverse-Fabius endpoint analysis. Equation (66) offers a second route through the large- s saddle of $B(s)$.

15.3 Autocorrelation arithmetic

The squared sinc product in (128) has a refinement mask obtained by squaring the original mask. It is reasonable to expect a dyadic arithmetic theory for $R(a)$ analogous to that of $\text{up}(a)$, but with triangular rather than uniform digit laws.

Conjecture 15.3 (Dyadic autocorrelation arithmetic). *For every dyadic rational $a \in [-2, 2]$, the value $R(a)$ belongs to the field generated by the Fabius dyadic values and, in fact, is rational after the natural power-of-two normalization determined by the spline rank. Its odd-prime denominators should be governed by products of factors $2^{2^m} - 1$ with multiplicities reflecting the squared mask.*

Even a weaker exact recurrence for $R(k/2^m)$ would be useful: it would turn shifted products of up into arithmetic data and would provide certified values of $\| \text{up} \|_2^2 = R(0)$.

15.4 Fractional operator spectrum

Equation (50) suggests studying the dilation–Volterra operator algebra generated by

$$(Df)(x) = f(x/2), \quad (V_\alpha f)(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt.$$

The Fabius function is a cyclic vector for a noncommutative relation between these operators. Promising questions include:

- meromorphic continuation of $J_{\alpha,\nu}$ jointly in (α, ν) after canonical endpoint subtractions;
- spectral decompositions of the dilation generator in Mellin coordinates;
- base- b analogues, replacing the dyadic triangular exponents by b -adic quadratic forms;
- a fractional analogue of the signed derivative hierarchy (7).

15.5 Mellin zeros, growth, and acceleration

The Newton series (61) can be evaluated entirely from rational moments. Its convergence is superalgebraic but not generally geometric. The endpoint phase suggests splitting d_n into a smooth saddle part and a tiny periodic residual, then applying Euler or Nørlund acceleration to the Newton basis. This may reveal:

- the order and indicator function of the entire function $D(z)$;
- zero patterns of D , $1 - D$, and the regularized Mellin transforms;
- functional relations between those zeros and the zeros of the Laplace product B ;
- certified high-precision negative moments, inaccessible to direct quadrature.

15.6 Orthogonal-polynomial arithmetic

The rational coefficients b_n in (134) deserve a separate arithmetic study. General orthogonal-polynomial theory predicts interval-universal limiting behavior because the density is positive on $(0, 1)$, but the superflat endpoints may leave a slow, log-periodic signature in the convergence rate. Concrete problems are:

- determine 2-adic and odd-prime valuations of b_n ;
- derive a nonlinear recurrence directly from the moment recurrence (52);
- identify the first log-periodic term in the large- n Jacobi asymptotics;
- connect Padé approximants of S to exact dyadic evaluations of F .

16 Novelty ledger and formalization roadmap

Result	Status in audit	Main proof mechanism
$\int_0^x F = F(x/2)$ and integer iterates	baseline, partly implicit	Fabius differential dilation and normalization
Finite formula (33)	not located explicitly	repeated integration by parts with shrinking argument
Complex series (39)	new relative to audit	exact recurrence plus endpoint-flat factorial bound
Newton–Mellin formula (61)	new relative to audit	dyadic endpoint/moment identity at $x = 1$
Fractional recurrence (50)	new relative to audit	Riemann–Liouville kernel and one integration by parts
Local inverse primitive (85)	not located explicitly	quantile substitution followed by Stieltjes integration by parts
All inverse derivatives (92)	standard inverse calculus, new packaging	differential-polynomial recurrence preserving $F^{r^{2n-1}}$
Sharp L^q threshold (97)	not located explicitly	change of variables and infinite endpoint flatness
Refined inverse primitive (106)	new consequence of repository asymptotics	elasticity identity and Watson expansion
Stieltjes/logarithmic transforms	new systematic synthesis	master CDF–survival and quantile identities
Gini/autocorrelation formulas	new synthesis	order statistics and Fourier representation of absolute distance
Rational Jacobi parameters	not located explicitly	rational moments, symmetry, and Gram–Schmidt

The following sequence is realistic for Lean formalization:

1. formalize A_p for $p \in \mathbb{N}_0$, prove (32), and derive the finite sum by induction;
2. formalize the n th Volterra primitive (45) using finite binomial expansion;
3. define $D(z)$ first on real exponents, prove the moment recurrence and Mellin identities, then add complex analyticity;
4. prove the finite Newton identities at integer z before introducing normal convergence of (61);
5. formalize the inverse local primitive and the Q_n recurrence, which require only one-dimensional calculus;
6. treat L^q divergence using the existing endpoint-flat lemmas;

7. postpone the refined periodic asymptotics until derivative-controlled error bounds are available in the formal library.

17 Conclusion

The Fabius–Rvachev system is unusually well suited to integration: differentiation doubles the argument, while normalized integration halves it. Once this elementary observation is combined with endpoint flatness, it yields more than the familiar primitive $F(x/2)$. Polynomial weights produce finite dyadic sums; arbitrary complex weights produce normally convergent dyadic series; the endpoint values turn those series into an entire Newton interpolation of complex moments; and quantile substitution transfers the same structure to the inverse Fabius function.

The transform picture is similarly unified. Mellin transforms are controlled by the entire moment function D , Laplace transforms by the product B , Fourier transforms by the lacunary sinc product, and Stieltjes transforms by a rational-moment Jacobi matrix. Products of F become order-statistic gaps, while shifted products of up become autocorrelations of the squared sinc product. The resulting formulas are exact, computationally testable, and closely connected to the repository’s existing themes of Thue–Morse cancellation, q -binomial arithmetic, Bernoulli/Bell expansions, and Lambert- W endpoint inversion.

The strongest immediate research opportunities are to formalize the complex-weight series and Newton interpolation, prove derivative-stable all-orders inverse asymptotics, develop the arithmetic of the rational Jacobi parameters and autocorrelation values, and use the Mellin–Laplace bridge to understand negative complex moments.

A Selected exact formulas at a glance

For convenience, the central formulas are collected here:

$$\begin{aligned}
 I_0^n F(x) &= 2^{\binom{n}{2}} F(x/2^n), \\
 \int_0^x t^p F(t) \, dt &= \sum_{j=0}^p (-1)^j \frac{p!}{(p-j)!} 2^{\binom{j+1}{2}} x^{p-j} F(x/2^{j+1}), \\
 A_\nu(x) &= \sum_{j \geq 0} (-1)^j \nu^j 2^{\binom{j+1}{2}} x^{\nu-j} F(x/2^{j+1}), \\
 \mathcal{M}_F(z) &= \frac{1 - D(z)}{z}, \quad \mathcal{M}_{\text{up},+}(z) = \frac{D(z)}{z}, \\
 D(z) &= 1 - z \sum_{j \geq 0} (-1)^j z - 1^j \frac{d_{j+1}}{(j+1)!}, \\
 B(s) &= \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \\
 \int_0^y G(t)^p \, dt &= y G(y)^p - p A_{p-1}(G(y)), \\
 G^{(n)}(y) &= \frac{Q_n[F'](G(y))}{F'(G(y))^{2n-1}},
 \end{aligned}$$

$$S(w) = \int_0^1 \frac{dy}{w - G(y)} = \sum_{n \geq 0} \frac{d_n}{w^{n+1}}.$$

B Bibliographic and repository notes

The repository was inspected through its live GitHub directory pages and raw TeX views on 27 August 2026. Because the draft and archive trees contain superseded copies, an archived duplicate was used to identify earlier claims but was not counted as an independent source. The accompanying report is self-contained and does not require the repository to compile.

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